

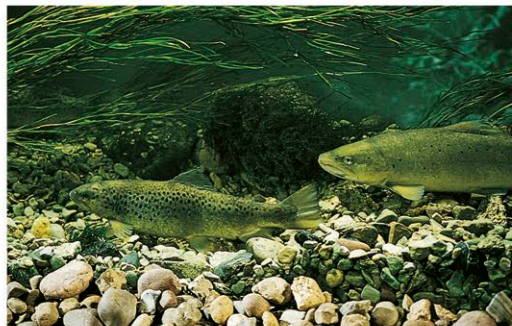
Life in freshwater

Freshwater is an essential resource for all land-dwelling animals, but it is also an important habitat in its own right. Freshwater habitats vary immensely, from temporary pools to giant lakes, and from tiny streams to rivers thousands of miles long. As a result, the problems that animals must overcome in order to survive in their environment can be very different from one freshwater habitat to another. Strong currents, periodic drought, and intense competition for food are some of the difficulties for which animals must find solutions.

Staying in place

Animals living in fast-flowing rivers face a constant battle with the current, and they deal with it in one of 2 main ways. The first is to avoid the problem by staying close to the riverbed, where the current is relatively slow. Many invertebrates, such as stonefly and mayfly larvae, never venture into open water; and to improve their staying power they often have a flat profile, so that if they position themselves with their heads facing upstream, the current presses down on their backs, helping them to stay in place. Dippers are the only songbirds that feed underwater, and can stay submerged even when moving against the current. They use their flapping wings to keep themselves in position, while their feet grip the substrate of the river bed.

The second solution is to compete with the current by swimming against it. Species that cannot avoid the current match their swimming speed to the water flow, enabling them to stay in place—and they keep swimming even when they are asleep.



COPING WITH THE CURRENT

Brown trout live in cool, fast-flowing streams, where it takes a lot of energy simply to stay in place. Trout can hold their own because fast-moving water is usually rich in oxygen—exactly what the fish need to keep their muscles working.

Migration

In addition to supporting freshwater residents, rivers and lakes receive visitors from the sea. These are migratory fish, which divide their time between fresh- and saltwater habitats. Anadromous species, such as salmon, breed in rivers but spend most of their adult lives offshore. Catadromous species, such as some eels, do exactly the opposite: they live in freshwater, but swim out to sea to breed.

For most fish, freshwater makes a much safer nursery than the ocean, and this justifies their long journey upriver to lay their eggs. However,

because the food supply in freshwater is limited, the young eventually make for the much more dangerous, but also more fruitful, environment of the open sea. The advantages of this are apparent where some, but not all, members of a species migrate: those that swim out to sea usually grow much bigger than those that stay behind.

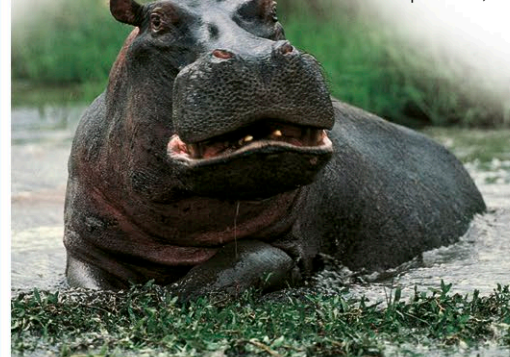
Like migrating birds, migrating fish are often remarkably accurate in pinpointing the place of their birth, returning there to spawn even though it might mean an inland journey of over 1,500 miles (2,500 km). Each river has its own characteristic chemical fingerprint and, with their acute sense of smell, migratory fish are able to identify the estuary that they left as young fish. By monitoring the scent of the water as they progress upriver, they home in on their own spawning ground.

Migrating fish often meet barriers to their progress. Salmon are famous for jumping waterfalls and rapids, while eels tackle obstacles by slithering around them over land. Eels usually do this after dark, in damp conditions, when they can survive out of water by breathing through their skin.

Living in and out of water

Most amphibians, such as frogs and toads, develop in freshwater habitats and then, as they approach adulthood, take up life on land, returning to water to breed. However, insects evolved amphibious lifestyles long before the first true amphibians appeared. Today, almost every patch of freshwater, from the smallest puddle to the largest lake, is inhabited by insects. Mosquito larvae feed on microscopic freshwater life, whereas the larvae of dragonflies and damselflies stalk larger prey, catching it with a set of extensible jaws called a mask. These insects leave the water when they become adult, but water beetles and bugs remain in it for life, although their ability to fly makes it easy for them to spread from one pool to another.

Some larger animals have developed lifestyles that straddle water and land. Snakes are good swimmers, and a number of species,



A DAYTIME REFUGE

Hippos lounge in rivers and lakes by day and emerge at night, sometimes traveling over 9 miles (15 km) to find good grazing. Their skin is thin and almost hairless, but it exudes a special secretion that helps to protect them from the daytime sun.

JOURNEY'S END

Most migratory fish travel upriver to breed several times during their lives. Sockeye salmon, however, almost always die once the adults have spawned. Sockeyes that are ready to breed typically develop hooked jaws and humped backs.

